

Bluestock Mutual Fund Analytics

Capstone Project — Data Analyst Internship

1. Project Overview

This project builds an end-to-end Mutual Fund Analytics Platform analyzing 40 Indian mutual fund schemes across 10 fund houses. The dataset includes 46,000 NAV records, 32,778 investor transactions, AUM data, SIP inflows, and portfolio holdings from 2022 to 2026.

2. Methodology

Data Source:

- 10 CSV datasets provided by Bluestock Fintech
- Live NAV data fetched from mfapi.in API

Data Cleaning Steps:

- Fixed 12 missing values in SIP yoy_growth_pct column
- Converted all date columns to datetime format
- Verified zero duplicate rows across all datasets
- Validated all 40 AMFI codes match between fund_master and nav_history

Tools Used:

- Python, Pandas, NumPy — data processing
- Matplotlib, Seaborn — visualizations
- SQLite — database storage
- Streamlit — interactive dashboard
- ReportLab, python-pptx — report generation

3. Dataset Summary

File	Rows	Description
fund_master	40	Scheme details, categories, risk grades
nav_history	46,000	Daily NAV prices 2022-2026
investor_transactions	32,778	Buy/sell transactions
benchmark_indices	8,050	NIFTY50 daily values
portfolio_holdings	322	Stock holdings per fund

4. Key Visualizations

Chart 1: Top 5 Funds by CAGR

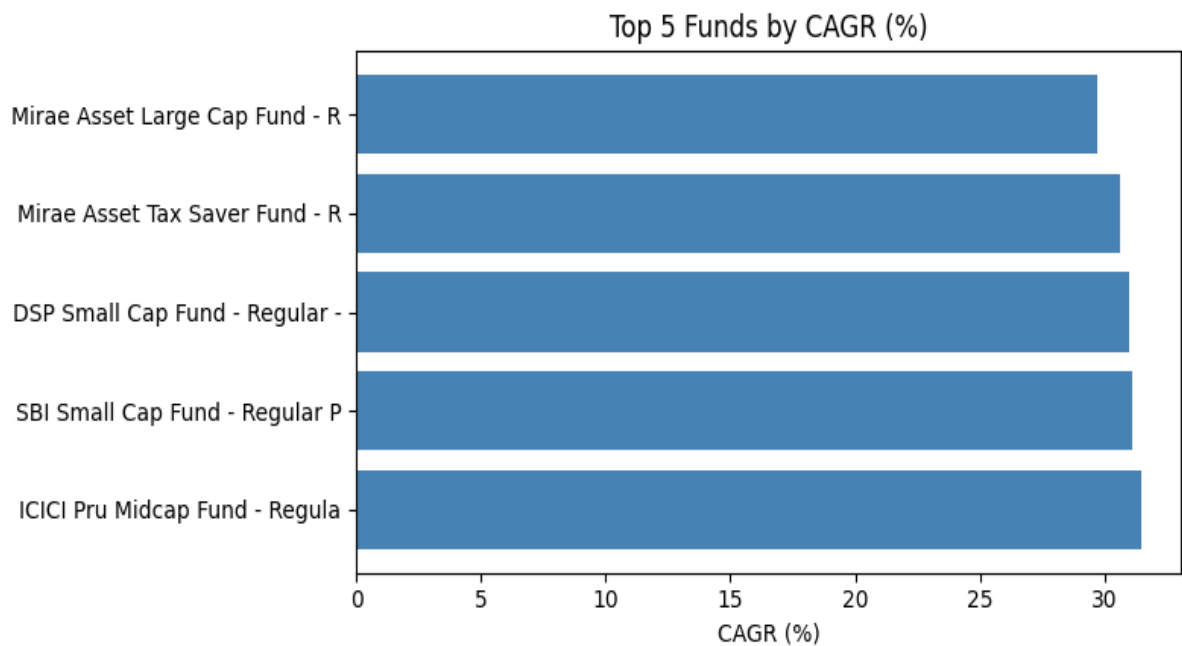


Chart 2: SIP Inflow Trend

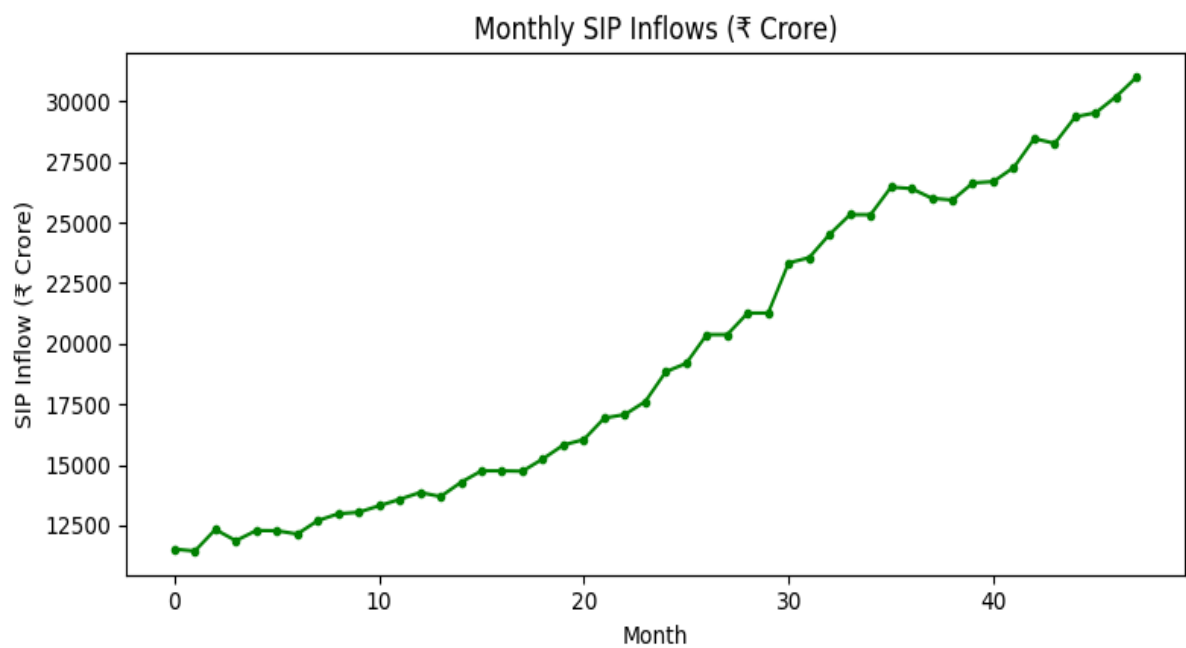
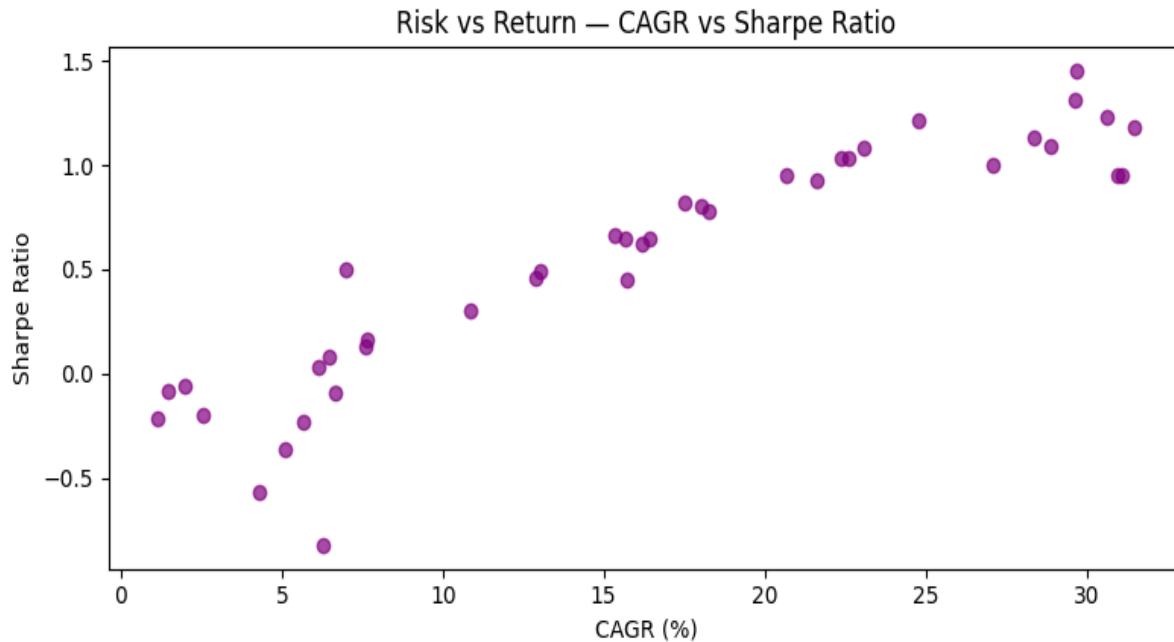


Chart 3: Risk vs Return



5. Top 5 Funds by CAGR

Scheme Name	CAGR %	Sharpe	VaR 95%
ICICI Pru Midcap Fund - Regular - Growth	31.48%	1.18	-1.89%
SBI Small Cap Fund - Regular Plan - Grow	31.1%	0.95	-2.45%
DSP Small Cap Fund - Regular - Growth	30.97%	0.95	-2.35%
Mirae Asset Tax Saver Fund - Regular - G	30.64%	1.23	-1.71%
Mirae Asset Large Cap Fund - Regular - G	29.71%	1.45	-1.36%

6. Recommendations

For Conservative Investors (Low Risk):

- ICICI Pru Liquid Fund — CAGR 6.97%, stable returns
- Kotak Liquid Fund — Low risk, consistent growth
- SBI Magnum Gilt Fund — Government backed, very safe

For Aggressive Investors (High/Very High Risk):

- ICICI Pru Midcap Fund — CAGR 31.48%, Sharpe 1.18
- Mirae Asset Tax Saver — CAGR 30.64%, tax benefits
- SBI Small Cap Fund — CAGR 31.10%, high growth

7. Conclusion

The analysis demonstrates that Indian mutual funds have delivered exceptional returns over 2022-2026. Mid and small cap funds outperformed with 28-31% CAGR. The fund recommender system helps investors choose the right fund based on their risk appetite. The Streamlit dashboard provides interactive exploration of all metrics and trends.